



Madurai Smart Citizen Service Nodes Stream

Municipal Citizen Grievance Analytics

Empowering the Madurai Corporation with cognitive real-time semantic processing. Raw spoken local dialects and unstructured complaints (Tamil & English mixed prose) are converted instantly into structured, prioritized records optimized for GIS dispatch and BigQuery pipeline analytics.

+ Active Grievance Simulator

CLICK ACTIVE NODES EXCERPT TO PRE-FILL:

- Pothole Risk (Mattuthavani)
- Garbage Dump (Periyar)
- Dark Junction (Goripalayam)
- Water Leak (Arappalayam)
- Stray Dog Menace (Anna Nagar)

1. CHOOSE TARGET ZONE

GIS BOUNDS ACTIVE

Mattuthavani Zone

2. RAW CITIZEN DIALECT PAYLOAD

0 chars

Type or click the microphone to speak regional language prose (mixed English and Tamil) with telephone numbers or PII info to



VOICE LANG

TOTAL INGESTED

5 100% INGEST



Grievance ledger table rows

CRITICAL ACTIONS

4 80% SEVERE



Severity score metric 4 or 5

PII SAFEGUARD

2 AUTO SCRUBBED



Confidential records masked

BQ SYNC RATE

100% REAL-TIME



Zero queue backlog lag

CATEGORIZED RESOLUTION UNITS DISTRIBUTION

Roads & Highways

1 28%

Sanitation & Waste

1 28%

Electrical & Streetlig...

1 28%

Water Supply

1 28%

Public Safety

1 28%

Active Grievance Ledger Matrix

LEDGER COUNT: 5 OF 5 ROWS

COMPLAINT ID	TARGET ZONE	EXTRACTED CATEGORY	TRANSLATED ENGLISH TEXT	SEVERITY INDEX	TARGET INGESTION STATUS	ACTIONS



Madurai Municipal AI Voice Node

LINK ESTABLISHED - SPEAK TAMIL / ENGLISH

ESC

Voice assistant link established. Start speaking...

AUDIO DECODER: 16kHz PCM_IN | 24kHz PCM_OUT



TERMINATE CONNECTION



Madurai Municipal AI Voice Node

● LINK STANDBY

ESC

Voice assistant link established. Start speaking...

Dialogue assistant link disconnected.

AUDIO DECODER: 16kHz PCM_IN | 24kHz PCM_OUT



ESTABLISH VOICE FEED LINK

Veo Video Animator Workspace

veo-3.1-fast

Generate hyper-realistic civic hazard animations to support smart public works simulations using Google Veo Image-to-Video models.

BASE IMAGE INPUT (STARTING FRAME)

 Pothole & Road

 Sanitation & Waste

Drag & drop or click to upload target frame photo...

MOTION ANIMATION PROMPT

Describe how the assets or particles in the municipal image should animate...

VIDEO ASPECT RATIO

LANDSCAPE (16:9)

PORTRAIT (9:16)

 GENERATE MOCK VEO VIDEO SEQUENCE



Awaiting Simulation Build

Load a municipal road or water line photo on the left, describe the motion, and trigger the compilation sequence.

VEO ENGINE RUNTIME CONNECTED

Close Workspace



- Pothole Risk (Mattuthavani)
- Garbage Dump (Periyar)
- Dark Junction (Goripalayam)
- Water Leak (Arappalayam)
- Stray Dog Menace (Anna Nagar)

1. CHOOSE TARGET ZONE

GIS BOUNDS ACTIVE

Mattuthavani Zone

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(mixed English and Tamil) with
telephone numbers or PII info to
see auto-redaction...



VOICE LANG
தமிழ் EN

PIPELINE ENGINE CONFIG

- Scrub PII & Telephone
- Semantic Translation
- Google Search Grounding
- Google Maps Grounding



INGEST & PROCESS PAYLOAD SEQUENCE



CATEGORIZED RESOLUTION UNITS DISTRIBUTION

Roads & Highways

1 28%

Sanitation & Waste

1 28%

Electrical & Streetlig...

1 28%

Water Supply

1 28%

Public Safety

1 28%

Active Grievance Ledger Matrix

LEDGER COUNT: 1 OF 5 ROWS



perl

ZONE:

All Zones

CAT:

All Categories

SEV:

All Severities

COMPLAINT ID	TARGET ZONE	EXTRACTED CATEGORY	TRANSLATED ENGLISH TEXT	SEVERITY INDEX	TARGET INGESTION STATUS	ACTIONS
MDU-VOICE-83110	Periyar	Sanitation & Waste	Complete trash and garbage accumulation opposite... Periyar Bus Stand. Offensive	4	SUCCESS_LOGGED	



- Pothole Risk (Mattuthavani)
- Garbage Dump (Periyar)
- Dark Junction (Goripalayam)
- Water Leak (Arappalayam)
- Stray Dog Menace (Anna Nagar)

1. CHOOSE TARGET ZONE

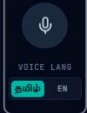
GIS BOUNDS ACTIVE

Mattuthavani Zone

2. RAW CITIZEN DIALECT PAYLOAD

0 chars

Type or click the microphone to
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telephone numbers or PII info to
see auto-redaction...



PIPELINE ENGINE CONFIG

- Scrub PII & Telephone ☒
- Semantic Translation ☒
- Google Search Grounding ☐
- Google Maps Grounding ☐

INGEST & PROCESS PAYLOAD SEQUENCE

CATEGORIZED RESOLUTION UNITS DISTRIBUTION

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Sanitation & Waste

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Electrical & Streetlig...

1 28%

Water Supply

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Public Safety

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Active Grievance Ledger Matrix

LEDGER COUNT: 1 OF 5 ROWS

Q peri

ZONE:

All Zones

CAT:

All Categories

SEV:

All Severities

COMPLAINT ID	TARGET ZONE	EXTRACT CATEGORY	ENGLISH TEXT	SEVERITY INDEX	TARGET INGESTION STATUS	ACTIONS
MDU-VOICE-83118	Periyar	Sanitation & Waste	and garbage opposite...nd. Offensive	4	SUCCESS_LOGGED	



- Pothole Risk (Mattuthavani)
- Garbage Dump (Periyar)
- Dark Junction (Goripalayam)
- Water Leak (Arappalayam)
- Stray Dog Menace (Anna Nagar)

1. CHOOSE TARGET ZONE

GIS BOUNDS ACTIVE

Mattuthavani Zone

2. RAW CITIZEN DIALECT PAYLOAD

0 chars

Type or click the microphone to speak regional language prose (mixed English and Tamil) with telephone numbers or PII info to see auto-redaction...



VOICE LANG

தமிழ்

EN

PIPELINE ENGINE CONFIG

Scrub PII & Telephone



Semantic Translation



Google Search Grounding



Google Maps Grounding



INGEST & PROCESS PAYLOAD SEQUENCE



CATEGORIZED RESOLUTION UNITS DISTRIBUTION

Roads & Highways

1 28%

Sanitation & Waste

1 28%

Electrical & Streetlig...

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Water Supply

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Public Safety

1 28%

Active Grievance Ledger Matrix

LEDGER COUNT: 1 OF 5 ROWS



peri

ZONE:

All Zones

CAT:

All Categories

SEV:

All Severities

COMPLAINT ID

TARGET ZONE

EXTRACTED CATEGORY

TRANSLATED ENGLISH T

TARGET INGESTION STATUS

ACTIONS

MDU-
VOICE -
83118

Periyar

Sanitation & Waste

Complete trash and gar
accumulation opposite
Periyar Bus Stand. Offe

Roads & Highways
Sanitation & Waste
Electrical
Water Supply
Public Safety

SUCCESS_LOGGED





Pothole Risk (Mattuthavani)

Garbage Dump (Periyar)

Dark Junction (Goripalayam)

Water Leak (Arappalayam)

Stray Dog Menace (Anna Nagar)

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GIS BOUNDS ACTIVE

Mattuthavani Zone

2. RAW CITIZEN DIALECT PAYLOAD

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VOICE LANG
தமிழ் EN

PIPELINE ENGINE CONFIG

Scrub PII & Telephone



Semantic Translation



Google Search Grounding



Google Maps Grounding



INGEST & PROCESS PAYLOAD SEQUENCE



COGNITIVE TRANSLATION REPORT

NODE_3000_INGRESS

CATEGORIZED RESOLUTION UNITS DISTRIBUTION

Roads & Highways

1 20%

Sanitation & Waste

1 20%

Electrical & Streetlig...

1 20%

Water Supply

1 20%

Public Safety

1 20%

Active Grievance Ledger Matrix

LEDGER COUNT: 1 OF 5 ROWS



peri

ZONE:

All Zones

CAT:

All Categories

SEV:

All Severities

COMPLAINT
ID

TARGET
ZONE

EXTRACTED
CATEGORY

TRANSLATED ENGLISH TEXT

SEVERITY
INDEX

TARGET
STATUS

MDU-
VOICE -
83110

Periyar

Sanitation
& Waste

Complete trash and garbage
accumulation opposite...
Periyar Bus Stand. Offensive

4

SUC

1 (Low)

2 (Minor)

3 (Moderate)

4 (High)

5 (Critical)



📍 Garbage Dump (Periyar)

📍 Dark Junction (Goripalayam)

📍 Water Leak (Arappalayam)

📍 Stray Dog Menace (Anna Nagar)

1. CHOOSE TARGET ZONE

GIS BOUNDS ACTIVE

Anna Nagar Park Zone

2. RAW CITIZEN DIALECT PAYLOAD

94 chars

Anna Nagar main street dog
menace near park kids cannot
play safely and bite fear 0452-
2534567



VOICE LANG
தமிழ் EN

PIPELINE ENGINE CONFIG

🔒 Scrub PII & Telephone



🔗 Semantic Translation



🌐 Google Search Grounding



📍 Google Maps Grounding



INGEST & PROCESS PAYLOAD SEQUENCE



COGNITIVE TRANSLATION REPORT

NODE_3000_INGRESS

CATEGORIZED RESOLUTION UNITS DISTRIBUTION

Roads & Highways

1 20%

Sanitation & Waste

1 20%

Electrical & Streetlig...

1 20%

Water Supply

1 20%

Public Safety

1 20%

📖 Active Grievance Ledger Matrix

LEDGER COUNT: 1 OF 5 ROWS

🔍 peri

ZONE:

All Zones

CAT:

All Categories

SEV:

All Severities

COMPLAINT ID	TARGET ZONE	EXTRACTED CATEGORY	TRANSLATED ENGLISH TEXT	SEVERITY INDEX	TARGET INGESTION STATUS	ACTIONS
MDU-VOICE-83110	📍 Periyar	Sanitation & Waste	Complete trash and garbage accumulation opposite... Periyar Bus Stand. Offensive	🔴🔴🔴 4	🟢 SUCCESS_LOGGED	🗑️



PIPELINE ENGINE CONFIG

- Scrub PII & Telephone ☒
- Semantic Translation ☒
- Google Search Grounding ☐
- Google Maps Grounding ☐

INGEST & PROCESS PAYLOAD SEQUENCE

COGNITIVE TRANSLATION REPORT NODE_3000_INGRESS

ORIGINAL REGIONAL INPUT

"Anna Nagar main street dog menace near park
kids cannot play safely and bite fear 0452-
2534567"

PII SCRUBBING & TELEPHONE MASK

Redacted phone: "0452-2534567" safely
filtered.

SEMANTIC ENGLISH MAPPING

"There is a stray dog menace on Anna Nagar
Main Street near the park. Children are unable
to play safely due to the fear of dog bites."

CATEGORY: Public Safety

SEVERITY: 3/5

LIVE DATA STREAM LOGS

BIGQUERY INGEST CLEAR

83118

Periyar Bus Stand. Offensive

Grievance Ingestion Diagnostic Report

ID: MDU-VOICE-68413

RAW INPUT STREAM TRANSCRIPT (Tamil/English mix)

"Anna Nagar main street dog menace near park
kids cannot play safely and bite fear 0452-
2534567"

SEMANTIC MAP TRANSLATOR

There is a stray dog menace on Anna Nagar Main
Street near the park. Children are unable to play
safely due to the fear of dog bites.

PII GUARD REDACTION SCRUB LEVEL

"Anna Nagar main street dog menace near park
kids cannot play safely and bite fear
[REDACTED_CONFIDENTIAL]"

Phone number token detected and masked: "0452-2534567"

COGNITIVE SCHEMA MAP

GIS ZONE: Anna Nagar CLASSIFIER 95%
CONFIDENCE:

SEVERITY CODE: 3 / 5 LOG DATE: 07/05/2026,

Pushed to table: mdu_corp_ledger COMMIT_SUCCESS



V.O.I.C.E. AI v2.4-MADURAI

Voice-Optimized Ingestion & Cognitive
Extraction



LIVE VOICE ASSISTANT



VEO VIDEO ANIMATOR



MADURAI SECURE ZONE: 07/05/2026, 08:51:36 PM



BIGQUERY PIPELINE: ACTIVE

PII SCRUBBING & TELEPHONE MASK

Redacted phone: "0452-2534567" safely
filtered.

SEMANTIC ENGLISH MAPPING

"There is a stray dog menace on Anna Nagar
Main Street near the park. Children are unable
to play safely due to the fear of dog bites."

CATEGORY: Public Safety

SEVERITY:

3/5

LIVE DATA STREAM LOGS

BIGQUERY INGEST

CLEAR

Simulated real-time streaming insert pipeline targeting
BigQuery table `mdu_corp_ledger`.

TABLE_INSERT

08:51:22 PM AM

Target Table:

mdu_corp_ledger

Status:

SUCCESS_LOGGED

INSERTION ROW SCHEMA JSON:

```
{
  "insertId": "MDU-VOICE-68413",
  "json": {
    "grievance_id": "MDU-VOICE-68413",
    "location_zone": "Anna Nagar",
    "raw_payload_text": "Anna Nagar main street dog",
    "redacted_text_secure": "Anna Nagar main street",
    "extracted_category": "Public Safety",
    "english_translation": "There is a stray dog me",
    "severity_index": 3,
    "pii_detected": true,
    "redacted_phone": "0452-2534567",
    "confidence_score": 0.95,
    "offensive_words_detected": [],
    "citations": [],
    "audited_by": "V.O.I.C.E. AI Engine v2.4 (Gemini)"
  }
}
```